

IILM UNIVERSITY, GREATER NOIDA • SCHOOL OF COMPUTER SCIENCE & ENGINEERING

INTERNSHIP REPORT PRESENTATION • FINAL PROJECT EVALUATION

SMARTBRANCH 360

Enterprise Branch Network Design, Multi-Layer Security & Automated Assurance

Cisco-AICTE Virtual Internship Program 2026 | In Collaboration with Cisco Networking Academy

Vaibhav Raj Trivedi — Project Lead / Individual Project Developer

Roll Number: 2410030723 | Section: 3CSE15 | 5th Semester | Batch 2024–2028 | vaibhav.trivedi.cs28@iilm.edu

Internship Period: 15 May 2026 – 19 June 2026 | Coursework Completed: 16 June 2026

Candidate Profile & Leadership Role

Vaibhav Raj Trivedi

PROJECT LEAD

Roll Number: 2410030723

Section: 3CSE15 (5th Semester)

Department of Computer Science and Engineering
IILM University, Greater Noida, U.P.

Primary Project Responsibilities:

- Network architecture & topology modeling
- 802.1Q VLANs & Router-on-a-Stick design
- DHCP scope, DNS server & NAT/PAT edge
- Inbound extended ACL guest isolation policy
- Five-layer intentional fault diagnosis & recovery
- Custom Python network assurance checker (25/25)

Institutional Profile

- Student Name: Vaibhav Raj Trivedi
- Roll Number: 2410030723
- Section: 3CSE15 | Semester: 5th Semester
- Institute: IILM University, Greater Noida, U.P.
- Programme: B.Tech CSE, Batch 2024–2028

Contact & Verification

- Institutional Email: vaibhav.trivedi.cs28@iilm.edu
- Roll Number: 2410030723
- Section: 3CSE15
- Cisco NetAcad Learner: vaibhav-trivedi-cs28-iilm-edu
- Verified on Cisco Networking Academy Portal

Internship Domain

- Computer Networking & Network Reliability Engineering
- Hardware & IOS: Cisco 2911 ISR, Catalyst 2960-24TT
- Protocols: 802.1Q, DHCP, DNS, NAT/PAT, ACL, SSH v2
- Automation: Python 3.12 & PyYAML Assurance

Internship Timeline

- Formal Duration: 15 May 2026 – 19 June 2026
- Coursework Completed: 16 June 2026
- Instructor: Mr. Abhinav Raghav (Cisco Academy)
- Status: 100% Completed & Verified (3 Certifications)

Cisco-AICTE Virtual Internship Program 2026

AICTE Patronage

National Internship Framework

Conducted under the statutory patronage of the All India Council for Technical Education (AICTE), Ministry of Education, Government of India. The program bridges academic engineering curricula with hands-on enterprise technology domains.

Cisco Networking Academy

Industry Technology Curriculum

Delivered through the global Cisco Networking Academy platform. Provided comprehensive training in enterprise routing and switching, packet analysis, Cisco IOS syntax, and discrete-event network modeling using Cisco Packet Tracer 8.x.

IILM University Academy

Academic Mentorship & Lab

Facilitated by the School of Computer Science & Engineering at IILM University, Greater Noida. Led by Academy Instructor Mr. Abhinav Raghav, supporting structured laboratory exercises and project milestones.

VERIFIED CISCO NETWORKING ACADEMY COURSEWORK (COMPLETED 16 JUNE 2026)

- 1. Getting Started with Cisco Packet Tracer | Verification ID: 31b1dd72-5542-4c2c-b203-4fdd8eed62cb
- 2. Exploring Networking with Cisco Packet Tracer | Verification ID: 94b24045-9aa3-42bd-8d9e-7ef003ffc23a
- 3. Networking Essentials | Verification ID: a0227e1b-e2a3-4531-9b44-1dd41bb20ac0

Enterprise Branch Connectivity & Security Challenges

Operational Context: A modern enterprise branch office hosts four disparate user domains: corporate employees, untrusted visitor guests, an internal server farm, and administrative personnel. Operating on flat, unsegmented networks exposes the organization to severe operational vulnerabilities.

Broadcast Congestion

Flat Local Domain

Deploying all endpoints in a single subnet causes excessive ARP broadcasts and NetBIOS chatter, degrading switch fabric bandwidth and wireless capacity.

Lateral Security Threat

Guest Traffic Leakage

Visitor wireless endpoints on guest Wi-Fi reside on the same broadcast domain as corporate finance PCs and servers, risking packet sniffing and unauthorized access.

Configuration Drift

Manual IP Conflicts

Static manual IP assignment for expanding staff pools leads to duplicate IP addresses, erroneous default gateways, and frequent branch connectivity loss.

Assurance Blindspot

Manual Audit Overhead

Network engineers rely on manual ad-hoc CLI inspection. Silent misconfigurations and trunk pruning errors persist undetected until catastrophic outages occur.

Project Mandate: Model the branch in Cisco Packet Tracer, enforce 802.1Q VLAN isolation, Router-on-a-Stick, dynamic DHCP/DNS, NAT/PAT, and ACL guest isolation. Crucially, intentionally inject and dia

Core Engineering & Validation Objectives

1. Collapsed-Core Topology

Model a resilient enterprise branch topology connecting 8 user PCs, central server, AP, 2 Catalyst switches, and Cisco 2911 router to simulated ISP.

2. Layer 2 VLAN Segmentation

Configure 802.1Q VLANs (10 Employee, 20 Guest, 30 Server, 99 Management) with robust inter-switch gigabit trunking.

3. Router-on-a-Stick (RoAS)

Deploy 802.1Q subinterfaces on Router R1 (Gi0/0.10 - .99) as default gateways for controlled inter-subnet traffic forwarding.

4. Automated Dynamic Services

Build Cisco IOS DHCP pools with scope exclusions (.1-.20); deploy SRV1 intranet HTTP portal and authoritative DNS name resolution.

5. NAT/PAT & Internet Egress

Configure Port Address Translation (NAT Overload) on R1 Gi0/1 and default static routing to simulated Internet (8.8.8.8).

6. ACL Guest Isolation

Author and enforce inbound extended ACL 'GUEST_ISOLATION' on Gi0/0.20, blocking guest access to internal subnets while permitting WAN browsing.

7. Management Plane Hardening

Enforce SSH version 2 with RSA 1024-bit key encryption on dedicated management VLAN 99, completely disabling unencrypted Telnet.

8. Systematic Fault Simulation

Intentionally inject, diagnose, and remediate five multi-layer operational faults across Layer 2 switching, Layer 3 routing, DHCP, DNS, and NAT.

9. Automated Python Assurance

Develop python_checker.py to programmatically audit configurations and CLI outputs against YAML requirements, generating instant remediation syntax.

Collapsed-Core Topology & Device Roles

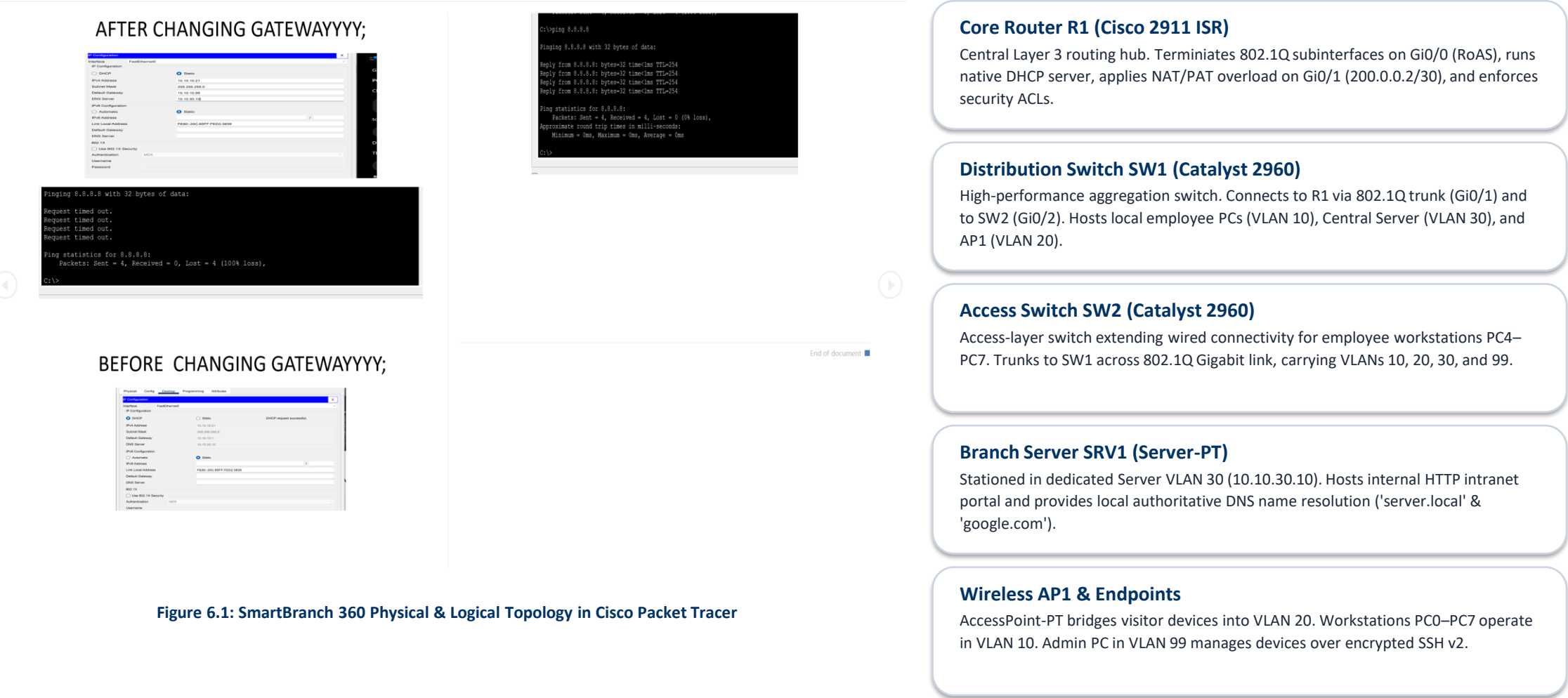


Figure 6.1: SmartBranch 360 Physical & Logical Topology in Cisco Packet Tracer

Hierarchical Subnetting & Gateway Schema

VLAN / Segment	Operational Purpose	Subnet CIDR	Default Gateway	IP Allocation Strategy
VLAN 10	Corporate Staff (PC0–PC7)	10.10.10.0/24 (255.255.255.0)	10.10.10.1 (R1 Gi0/0.10)	DHCP: .21–.254 (.1–.20 Reserved)
VLAN 20	Visitor Guests (AP1 Wi-Fi)	10.10.20.0/24 (255.255.255.0)	10.10.20.1 (R1 Gi0/0.20)	DHCP: .21–.254 (.1–.20 Reserved)
VLAN 30	Central Server Farm (SRV1)	10.10.30.0/24 (255.255.255.0)	10.10.30.1 (R1 Gi0/0.30)	Static: SRV1 @ 10.10.30.10/24
VLAN 99	Management Plane (SSH)	10.10.99.0/24 (255.255.255.0)	10.10.99.1 (R1 Gi0/0.99)	Static: SW1 (.2), SW2 (.3), Admin (.10)
WAN Link	Simulated ISP Transit	200.0.0.0/30 (255.255.255.252)	200.0.0.1 (ISP Gi0/0)	Static: R1 Gi0/1 @ 200.0.0.2/30

Deterministic Octet Mapping

The third octet of each private subnet corresponds directly to its 802.1Q VLAN ID (e.g., VLAN 10 -> 10.10.10.0/24). This provides immediate visual diagnostic clarity during packet tracing and CLI auditing.

Exclusion Pools & Zero Conflicts

Cisco IOS global exclusion commands ('ip dhcp excluded-address 10.10.x.1 10.10.x.20') reserve the lower 20 addresses in every subnet for gateways, switch SVIs, and servers, guaranteeing zero IP conflicts.

External Public Addressing

A point-to-point /30 subnet (200.0.0.0/30) conserves public IPv4 addresses on the WAN transit link. Simulated Internet target 8.8.8.8 resides on the ISP Loopback0 interface to validate edge routing.

Implementation Protocols & Multi-Layer Security

CORE NETWORKING TECHNOLOGIES

- **Cisco Packet Tracer 8.x:** Discrete-event simulation, packet inspection, and topology modeling.
- **Cisco IOS 15.x CLI:** Production command syntax for ISR routers and Catalyst switches.
- **IEEE 802.1Q Trunking:** VLAN encapsulation across gigabit links with explicit allowed lists.
- **Router-on-a-Stick (RoAS):** Logical subinterfaces (Gi0/0.x) performing Layer 3 inter-VLAN routing.
- **Dynamic Services (DHCP/DNS):** Cisco IOS DHCP server pools and SRV1 authoritative DNS mapping.
- **NAT/PAT (Port Address Translation):** Dynamic overload multiplexing internal 10.10.0.0/16 subnets to public WAN.

Inbound Extended ACL: GUEST_ISOLATION

Applied inbound on Router subinterface Gi0/0.20. Explicitly drops all guest IP packets destined for Employee (10.10.10.0/24), Server (10.10.30.0/24), and Management (10.10.99.0/24), while permitting outbound WAN web access ('permit ip ... any').

Dedicated Management Plane (VLAN 99)

Administrative device access is strictly isolated onto VLAN 99. Switch Switch Virtual Interfaces (SVIs) and router management IP reside outside user broadcast domains, preventing unauthorized management probing.

Cryptographic SSH Version 2 Hardening

Disabled unencrypted Telnet entirely. Enforced SSH version 2 with 1024-bit RSA asymmetric key pairs ('crypto key generate rsa modulus 1024'), local privilege 15 credentials, and 'transport input ssh' on VTY lines 0–4.

Layer 2 Broadcast & Trunk Isolation

Catalyst access ports configured with explicit switchport access modes. Inter-switch trunk ports on SW1 and SW2 prune unnecessary broadcast domains via strict 'switchport trunk allowed vlan' lists.

Automated Configuration Auditing & Remediation

1. Input Ingestion

Ingests structured YAML topology models & raw Cisco IOS show commands ('show run', 'show interfaces trunk').

2. Multi-Point Audit

Validates 8 dimensions: VLANs, trunk allowed lists, gateways, DHCP scopes, DNS, NAT, ACLs, and SSH.

3. Severity Tagging

Flags compliant rules as [PASS], minor drift as [WARNING], and network-impacting misconfigurations as [FAIL].

4. Auto Remediation

Dynamically generates the exact Cisco IOS CLI commands required to restore failing parameters to compliance.

ASSURANCE AUDIT SPECIFICATION MATRIX

- VLAN Audit: Verifies VLANs 10, 20, 30, 99 exist and active on switches
- Trunk Audit: Verifies 802.1Q encapsulation & allowed list on Gi0/1, Gi0/2
- Gateway Audit: Verifies 802.1Q subinterfaces Gi0/0.10-.99 on Router R1
- DHCP Audit: Verifies scopes EMPLOYEE, GUEST, SERVER & exclusions .1-.20
- DNS Audit: Verifies SRV1 daemon 'On' and A-records (server.local, google.com)
- NAT Audit: Verifies ACL 1 binding to Gi0/1 with 'overload' keyword
- ACL Audit: Verifies GUEST_ISOLATION deny internal / permit any rules
- SSH Audit: Verifies SSH v2, RSA 1024-bit key, and 'transport input ssh'

EMPIRICAL BENCHMARK PERFORMANCE

HEALTHY BASELINE SCORE: 100.0%

25 out of 25 automated checks evaluated as [PASS] with zero warnings or failures on the verified production baseline.

FAULT DETECTION RATE: 100.0% (5/5)

Correctly detected and isolated all five intentional multi-layer faults, outputting exact Cisco IOS remediation commands for immediate recovery.

Multi-Layer Fault Simulation & Systematic Remediation

Fault #	Layer / Domain	Observed Diagnostic Symptom	Root Cause Analysis	Verified Remediation Syntax
Fault 1	Layer 2 Switching	Endpoints in VLAN 20 cannot acquire DHCP leases; 100% loss to default gateway 10.10.20.1.	VLAN 20 pruned from SW1 Gi0/2 allowed list ('switchport trunk allowed vlan remove 20').	SW1(config-if)# switchport trunk allowed vlan add 20 Verified via 'show interfaces trunk'.
Fault 2	Layer 3 Host NIC	Workstation pings local subnet (10.10.10.2) OK, but fails to reach Server or Internet (8.8.8.8).	Host default gateway statically set to 10.10.10.254 instead of 10.10.10.1.	Host IPv4 GUI: Set Default Gateway = 10.10.10.1 (or toggle DHCP). Verified via 'ping 8.8.8.8'.
Fault 3	Service DHCP Scope	All VLAN 10 hosts simultaneously lose off-subnet access upon renewing their DHCP lease.	Administrative typo in router DHCP pool: 'default-router 10.10.10.99' configured on R1.	R1(dhcp-config)# default-router 10.10.10.1 Clients run 'ipconfig /renew'.
Fault 4	Application DNS Daemon	Pinging 8.8.8.8 works (100%), but browsing 'server.local' or pinging 'google.com' fails.	DNS daemon process stopped on branch server SRV1 (10.10.30.10).	SRV1 -> Services -> DNS -> Toggle 'On'. Verified domain ping resolves to 8.8.8.8.
Fault 5	Network NAT / PAT	Internal routing normal; internal hosts experience 100% timeout to simulated Internet 8.8.8.8.	NAT overload mapping statement removed from WAN Router R1.	R1(config)# ip nat inside source list 1 interface Gi0/1 overload Verified 'show ip nat stats'.

End-to-End Connectivity & Security Validation

FUNCTIONAL NETWORK VALIDATION MATRIX

- ✓ **Intra-VLAN Connectivity:** 100% SUCCESS (0% loss) between PC0-PC3 on SW1 and PC4-PC7 on SW2 within VLAN 10.
- ✓ **Inter-VLAN Routing (RoAS):** 100% SUCCESS. Corporate PCs in VLAN 10 route seamlessly to Server SRV1 in VLAN 30 via R1 subinterfaces.
- ✓ **DHCP Lease Acquisition:** 100% SUCCESS. Endpoints automatically obtain valid IP, subnet mask, default gateway, and DNS pointers upon booting.
- ✓ **Intranet & DNS Resolution:** 100% SUCCESS. 'server.local' resolves to 10.10.30.10 (HTTP portal loads); 'google.com' resolves to 8.8.8.8.
- ✓ **NAT/PAT Internet Egress:** 100% SUCCESS. Internal endpoints successfully ping 8.8.8.8 across WAN; 'show ip nat translations' confirms dynamic state table.
- ✓ **ACL Guest Isolation:** 100% ENFORCED. Guest PC pings to Employee PC (10.10.10.x) and Server (10.10.30.10) DROP, while pings to 8.8.8.8 SUCCEED.
- ✓ **Encrypted SSH v2 Access:** 100% VERIFIED. Admin PC on VLAN 99 connects via SSH v2 (RSA 1024-bit); unencrypted Telnet connections rejected.

PROGRAMMATIC ASSURANCE VERIFICATION

Automated Compliance Score

Status: 25 / 25 Checks PASS (100.0%)

Baseline configuration audited against requirements.yaml achieved 100% compliance across all 8 architectural dimensions.

Intentional Fault Detection

Status: 5 / 5 Fault Scenarios Flagged (100.0%)

python_checker.py successfully identified all 5 simulated faults, tagging each with exact [FAIL] severity indicators.

Actionable Remediation Engine

Status: 100% Automated CLI Scripting

For each detected fault, the checker dynamically generated the exact Cisco IOS syntax required to resolve the defect.

Verification Method

Status: Dual Ingestion Pipeline

Tested using both structured YAML topology declarations and live Cisco IOS show-command text captures ('show run', 'show interfaces trunk').

Project Leadership & Team Responsibilities

VAIBHAV RAJ TRIVEDI — PROJECT LEAD (Roll No: 2410030723, Section: 3CSE15)

- Network Architecture & Layout: Selected hardware models (Cisco 2911 ISR, Catalyst 2960-24TT, AP, Server) and planned inter-switch trunk links.
- Addressing & Subnetting: Engineered hierarchical 10.10.0.0/16 IPv4 plan, aligning subnets with VLAN IDs and allocating WAN 200.0.0.0/30 link.
- Layer 3 Routing & Services: Implemented 802.1Q subinterfaces on Router R1, native DHCP server pools with exclusions, and SRV1 DNS/HTTP services.
- Perimeter & Edge Security: Deployed Port Address Translation (NAT Overload) on R1 Gi0/1 and authored extended ACL 'GUEST_ISOLATION' on Gi0/0.20.
- Management Hardening: Configured SSH v2 with 1024-bit RSA keys on dedicated management VLAN 99, disabling unencrypted Telnet.
- Fault Simulation & Root Cause Analysis: Simulated 5 multi-layer faults, diagnosed symptoms via show commands, and verified recovery.
- Network Assurance Automation: Authored python_checker.py from scratch to parse configurations and audit compliance against requirements.yaml.
- Project Blueprints & Reporting: Authored full technical report, configuration listings, demonstration sequence, and repository documentation.

TEAM COLLABORATION & ACKNOWLEDGEMENT

While I served as the Project Lead holding end-to-end responsibility for the architectural design, configuration, testing, and Python assurance software, I gratefully acknowledge the contributions of my peers during testing and review:

Harsh Singh — Team Member

Assisted in endpoint connectivity test matrix execution and verification of intra-VLAN employee workstation communication.

Parveen — Team Member

Assisted in laboratory documentation review and verification of wireless access point guest onboarding test cases.

Key Competencies Gained & Technical Takeaways

Layer 2/3 Engineering Mastery

Gained practical mastery of IEEE 802.1Q trunking, native VLAN security, and Router-on-a-Stick subinterfaces, understanding how logical segmentation optimizes hardware efficiency.

Edge Security & Service Delivery

Designed and deployed native DHCP scope protection, dual-zone DNS name resolution, dynamic NAT/PAT overload, and named extended ACLs for source-proximate filtering.

Methodical Fault Troubleshooting

Developed a disciplined, layer-by-layer troubleshooting framework that isolates physical, data-link, routing, service, and application errors to minimize MTTR.

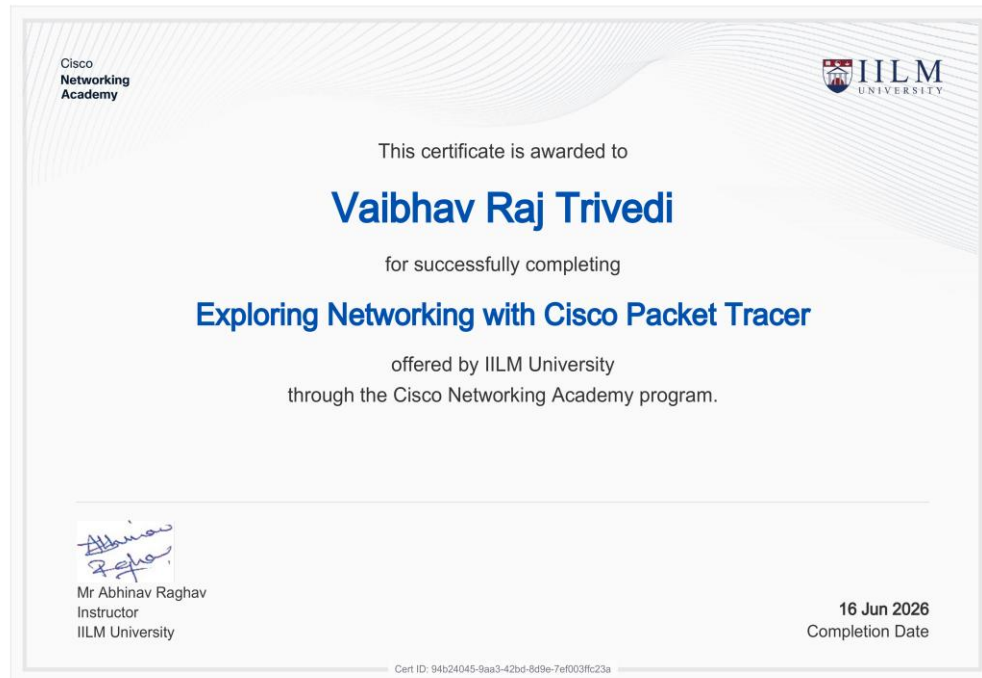
NetDevOps & Assurance Automation

Engineered software-driven network validation using Python 3.12 and PyYAML, bridging classical simulation with modern automated network reliability engineering.

CONCLUSION & IMPACT

The SmartBranch 360 project successfully demonstrates that combining hierarchical network segmentation, multi-layer perimeter security, and automated programmatic assurance produces an enterprise-ready, resilient branch architecture. The project met all functional milestones with 100% baseline operational connectivity, 100% guest isolation, 100% fault diagnosis recovery, and 25/25 automated Python assurance compliance.

Cisco Networking Academy Official Certifications



Exploring Networking with Cisco Packet Tracer

Instructor: Mr Abhinav Raghav | Date: 16 Jun 2026

Verification ID: 94b24045-9aa3-42bd-8d9e-7ef003ffc23a

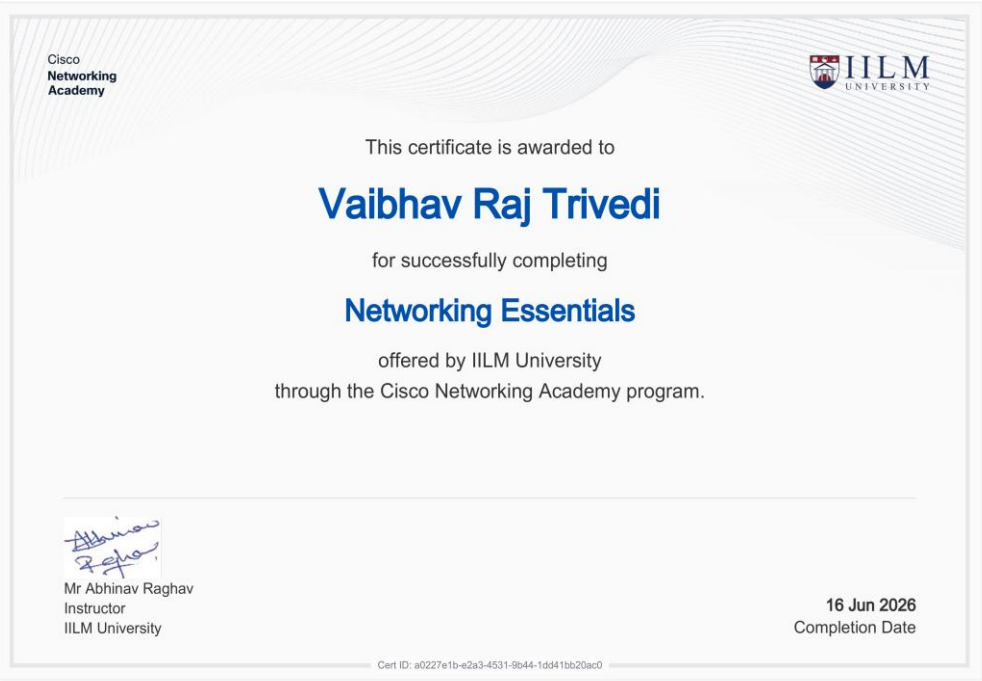


Getting Started with Cisco Packet Tracer

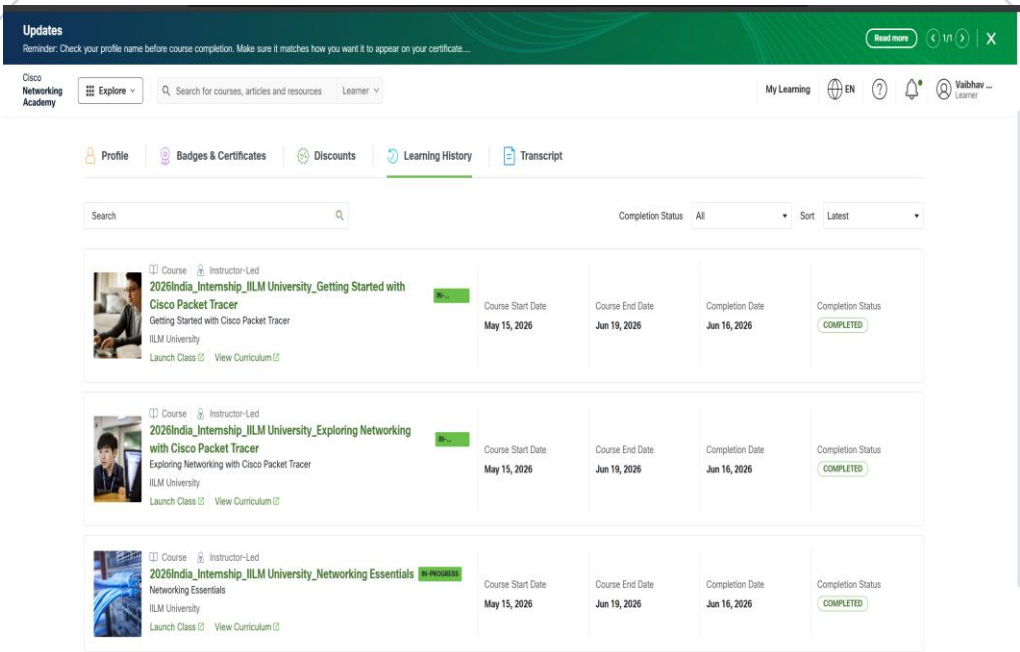
Instructor: Mr Abhinav Raghav | Date: 16 Jun 2026

Verification ID: 31b1dd72-5542-4c2c-b203-4fdd8eed62cb

Networking Essentials & NetAcad Enrollment Proof



Networking Essentials (Core Industry Certification)
Instructor: Mr Abhinav Raghav | Date: 16 Jun 2026
Verification ID: a0227e1b-e2a3-4531-9b44-1dd41bb20ac0



Official NetAcad Course Timeline & Verification Evidence
Cisco-AICTE Period: 15 May 2026 – 19 June 2026 | Completed: 16 June 2026
Enrolled under Instructor Mr. Abhinav Raghav, IILM University

Thank You

SMARTBRANCH 360

Enterprise Branch Network Design, Multi-Layer Security & Automated Assurance

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